NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Database Engineering CS2008 ■ Supplementary / Summer Sem 2022-23 ■ Pg 1 ■ FM 50 ■ 3 Hrs

■ Answer ALL questions. ■ All parts of a question MUST be answered together. ■ Mere answers without proper explanation will not fetch marks. ■ Variation in quality of answer will vary the secured marks.

1. Using Membership algorithm, check if $BD \rightarrow E$ is a redundant FD in $F = \{A \rightarrow B,\ C \rightarrow D,\ BD \rightarrow E,\ AE \rightarrow C\}$ [5]

2. *a.* What are different states of a transaction? What is a *serializable* schedule?
   b. Consider a relation R(ABCD) with FD set $\{D \rightarrow C,\ C \rightarrow B,\ B \rightarrow A,\ A \rightarrow D\}$. Derive how many non-trivial FDs are there in the closure of FD set in R?
   c. What is a dirty read problem? Explain with an example.
   d. What is *theta* join? What are properties of the operator *theta*?
   e. Find the all possible conflicts among T_1, T_2, T_3, and T_4 in S:
   $S : \{R_1(A),\ W_2(A),\ R_2(B),\ R_3(A),\ W_1(A),\ W_3(A),\ W_1(B),\ R_4(A),\ W_4(A)\}$
   f. Are vertical and horizontal partitioning commutative among themselves? [(1+1)+3+(1+1)+3+4+1]

3. *a.* What is condition for a relation to be in second normal form?
   b. Define weak entity with an example.
   c. For the schema R (A, B, C) in right side, and for given data therein, find how many lossless decomposition into two relations are possible where any of the resulting relations will not have less than or more than two attributes?
   d. What is a recursive relationship? Explain with an example.
   e. Show how a mandatory participation is shown in a min-max notation in ER diagram.
   f. Find the result of the division of R by $\pi_A(S)$ given below.

R(A, B, C)

A	B	C
a	6	7
b	3	4
c	8	4

R(A, B, C)

A	B	C
a_1	b_1	c_1
a_2	b_2	c_1
a_3	b_3	c_3
a_4	b_4	c_2
a_3	b_2	c_1
a_2	b_3	c_1
a_3	b_3	c_1
a_4	b_3	c_1

S(A,C)

A	C
a_1	c_1
a_2	c_1
a_3	c_3

[2+2+3+(1.5+1)+1+4.5]

4. *a.* What is the difference between DML and DCL? Explain with an example.
   b. Find the content of tables G,H,I,J,K,L,M,N,P from the following tables R and S:
   $G \leftarrow R \bowtie_{A=D} S$ $H \leftarrow R \ltimes_{A=D} S$ $I \leftarrow R \rtimes_{A=D} S$ $J \leftarrow R \lhd_{A=D} S$ $K \leftarrow R ⟗_{A=D} S$
   $L \leftarrow R \rhd_{A=D} S$ $M \leftarrow R ⟖_{A=D} S$ $N \leftarrow R ⟕_{A=D} S$ $P \leftarrow R * S$

R(A, B, C)

A	B	C
a_1	b_1	c_1
a_2	b_2	c_1
a_2	b_1	c_4
a_3	b_2	c_2
a_4	b_3	c_3

S(D, E, F)

D	E	F
a_2	e_1	f_1
a_1	e_2	f_3
a_1	e_2	f_2
a_2	e_4	f_4
a_3	e_2	f_1

   c. What is meant by *soundness* and *completeness* of Armstrong's axioms?
   d. Derive pseudotransitive rule using Armstrong's axioms. $[3 + (1 \times 9) + 1 + 2]$

■ BEST OF LUCK ■